



DEALER BEST PRACTICE SERIES

Hydraulic Backhoe -
Hammer to remove spool
and wedge retention
mechanism – GET

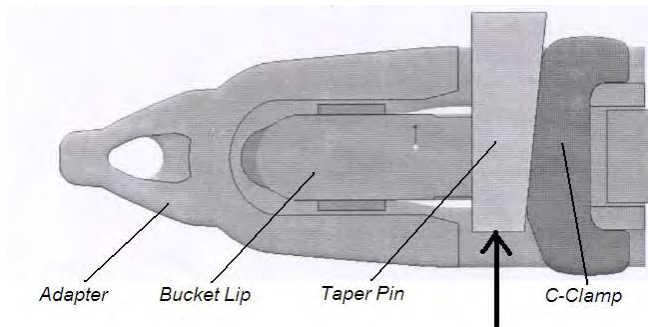
Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

Some specialized mining work still remains essentially manual. This is the case for the installation and removal of the spool and wedge retention system found on large mining Ground Engaging Tools (GET). Because the work is manual, there may be increased safety risk. The following is one dealer's suggestion for an improvement the dealer believes has reduced safety issues.



Current process - Unsafe

2.0 Best Practice Description

The use of a hydraulic hammer fitted with a wrist mount that can be orientated on a small backhoe attachment installed on the pit crew service truck has removed most of the sledge hammer work associated with the task of exchanging GET. This has resulted in reduced repair time for GET exchange and improved safety conditions of the task.

3.0 Implementation Steps

The customer identified the task of large GET removal as potentially dangerous. The physical strain associated with the extensive use of a sledge hammer, and the situation where pieces of metal dislodge from either the hammer or the wedge and become airborne projectiles.

The taper pin is installed from the top, and to remove it, it needs to be “unfrozen” with a hammer or else from the underside; then it can be removed manually from the top. This is the manufacturer's recommended removal procedure for this taper pin.

A team was formed to find alternate solutions. After evaluating what was available on the market and with GET suppliers, it was decided to have a team work on a local solution.

Trials were first made with a pneumatic hammer on a telescoping arm, but two downsides were identified: the large compressor needed (40cfm) and there are freezing problems associated with compressed air in sub-zero temperatures. The hydraulic hammer option was then reviewed, and because the service truck already had a hydraulic crane to maneuver the GET, few modifications were needed to add the new CAT BH160 backhoe attachment with the CAT H35D hammer on the truck's rear. The modifications included: fitting the backhoe on the truck's frame, upsizing the hydraulic tank on the truck, adding a hydraulic pump on the truck's Power Take Off (PTO).

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DATE
2/16/2015

CHG
NO
01

NUMBER
1006-1.05-1208

The backhoe comes as a complete attachment available for fitment on skid steer loaders.

The selected hydraulic positioner (shown in red) is made by Rototilt.

A radio frequency wireless remote control was added by a third party to add convenience mainly for the line of sight of the operator and keeping him at safe distance in case of possible airborne metal projections.

Only a few sharp blows is needed without much pressure; once loosened the taper pin can be removed by hand.

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A 45 degrees bracket was added between the Rototilt and the hammer to increase the angle of attack.



Another improvement could be a telescopic boom to increase the reach of the equipment.

4.0 Potential Benefits

Since the new equipment has been put in service, the only sledge hammer work still done is to replace one or two broken adapters. The above equipment is used for all major work. It has boosted productivity because human fatigue has been drastically reduced, and the number of safety issues related to sledge hammer usage has been substantially reduced.

5.0 Resources Required

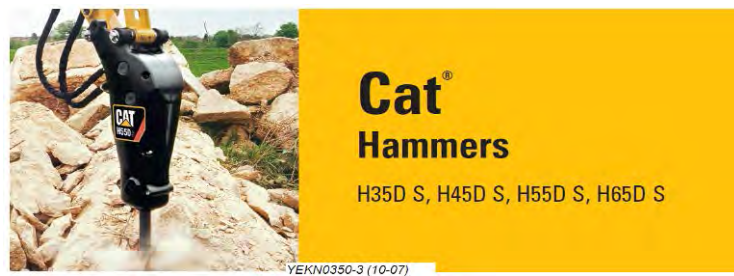
Management buy-in
Project manager
Mechanical engineer
Welder, mechanic

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6.0 Supporting Attachments / References

www.rototilt.com



7.0 Related Best Practices

8.0 Acknowledgements

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